

How to Read a Mathematical Paper as an Argumentative Essay

**A Method for Obtaining Digital Signatures
and Public-Key Cryptosystems**

R.L. Rivest, A. Shamir, and L. Adleman*



Key terms



- ❑ RSA - Names of the Authors “R.L. Rivest, A. Shamir, and L. Adleman”
- ❑ Encryption - A method to hide information by an established method
- ❑ Decryption - A method to uncover hidden information from an established method
- ❑ Public Key Cryptosystem - A system using a common directory of public keys for private encryption and decryption.

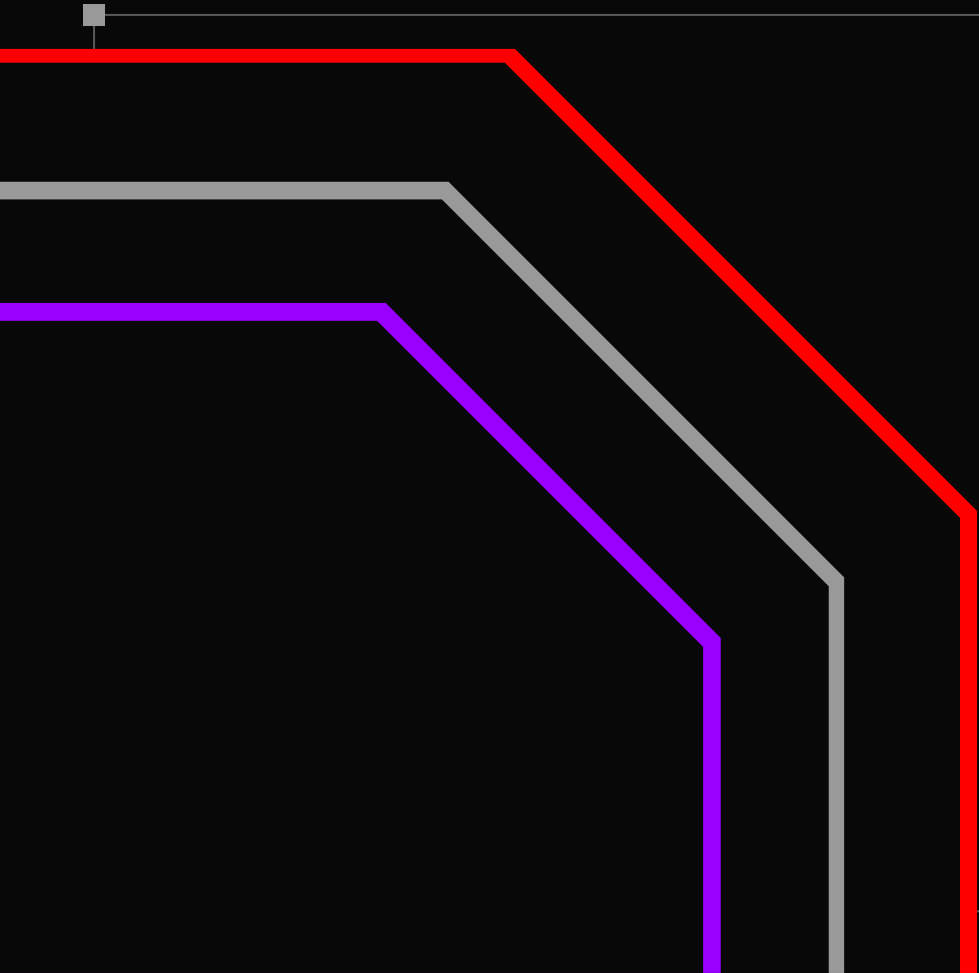


Key terms - Public Key Cryptosystem



- ❑ Public Key/Encryption Key - A public code containing both details to encrypt the message and what numeric key/number to use.
- ❑ Private Key - A private numeric key/number to decrypt the message.
- ❑ Signature - A numeric code attached to a document that could only be produced with such message and by such sender.
- ❑ RSA Cryptosystem - A public key cryptosystem developed by RSA.





Unit 1: Remember Two Reasons

1.1: See Who's Talking

1.2: Distinguish Major Sections

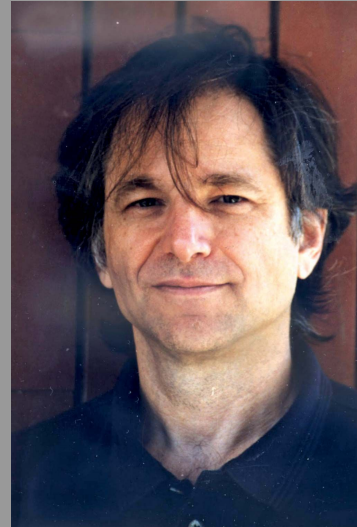
1.3: Look Up Technical Terms

1.4: Find the Main Claim

1.5: State the Main Claim
Accurately

1.1 The Authors

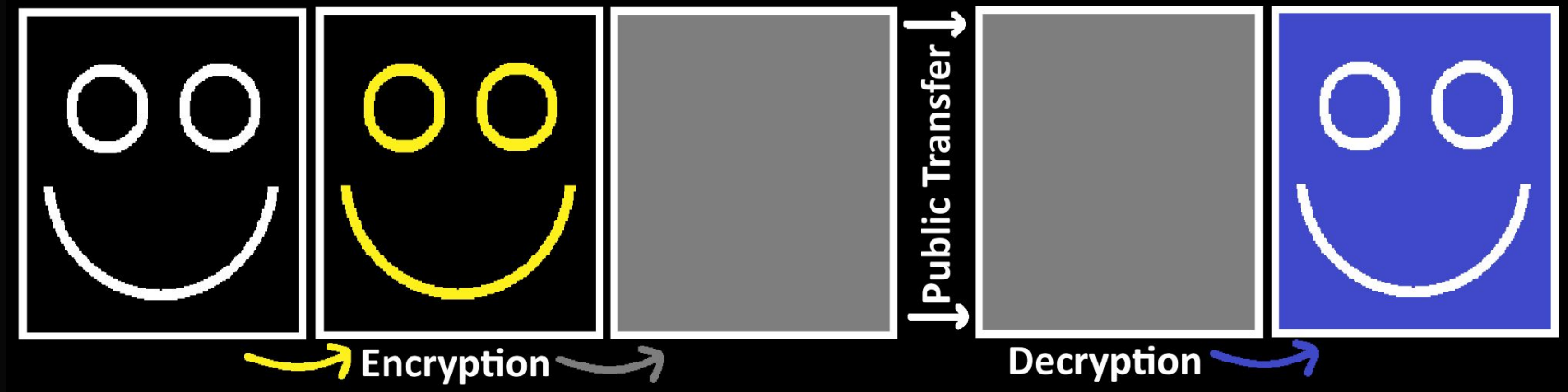
In this paper the speakers are R.L. Rivest, A. Shamir, and L. Adleman (RSA).



1.3 RSA Procedure

Lets say Alice wants to send an encrypted message to Bob but does not want anyone else understand.

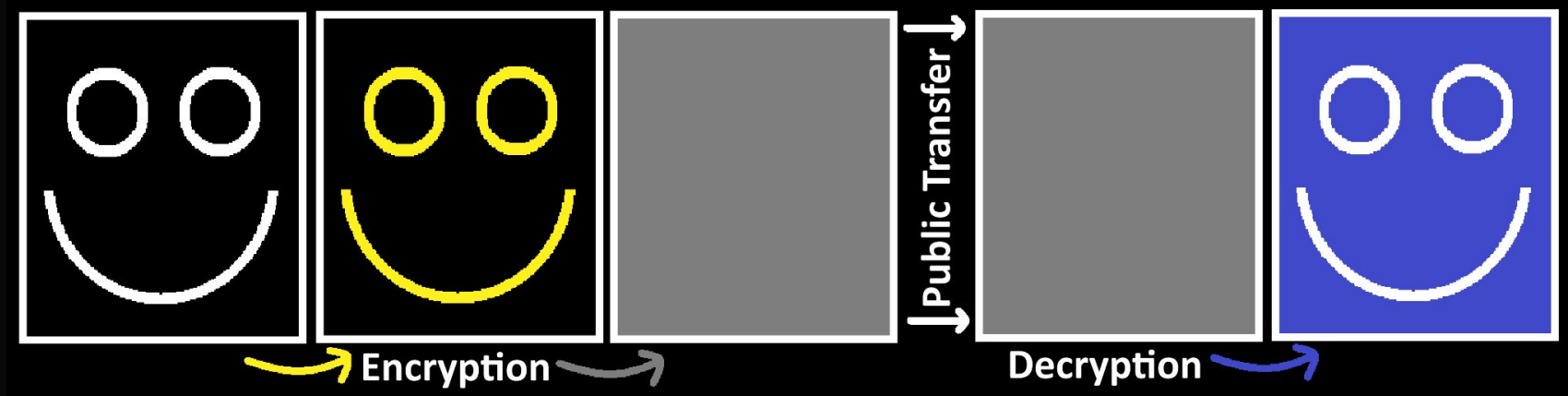
- ❑ First, Bob computes a private key d and public key (n, e) combination using RSA's algorithm.
- ❑ Next, Alice obtains Bob's and publicly available public key (n, e) and uses it to encrypt their message.



1.3 RSA Procedure

Lets say Alice wants to send an encrypted message to Bob but does not want anyone else understand.

- ❑ Alice then sends the encrypted message to Bob.
- ❑ After receiving Alice's message Bob decrypts it using his private key **!**.



1.5 Main Claim Revision

Main Claim: **"The RSA method for public-key cryptography is an acceptable and legitimate security technology."**

The RSA method for public-key cryptography ought to be regarded as acceptable and legitimate security technology.

Unit 2: Remember Two Reasons

2.1: Identify Two Reasons

2.2: Design a 10 Box Model

2.3: Find Missing Co-Premises

2.4: Collect Additional Reasons

2.1 Main Claim

Main Claim:

- The RSA method for public-key cryptography is an acceptable and legitimate security technology.

Why we believe this:

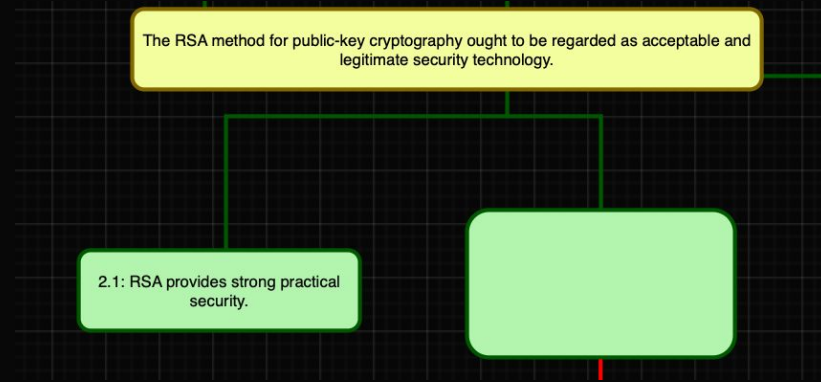
- The paper convince reader this mathematical method solves real world encryption problems.
- There are two main reasons of support that address the flaws of current encryption

2.1 Reason One - it's hard to break

RSA provides **strong practical security**.

Evidence:

- The security of RSA results from the **difficulty of factoring** the product of two large secret prime numbers. .
- Factoring has been studied by mathematicians for over 300 years; the fact that no efficient algorithm has been found to solve this problem shows that RSA is **difficult to reverse**.

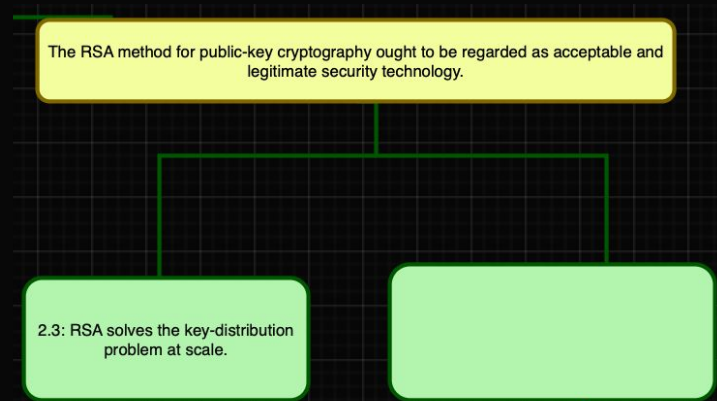


2.1 Reason Two- it's easier than Existing Methods

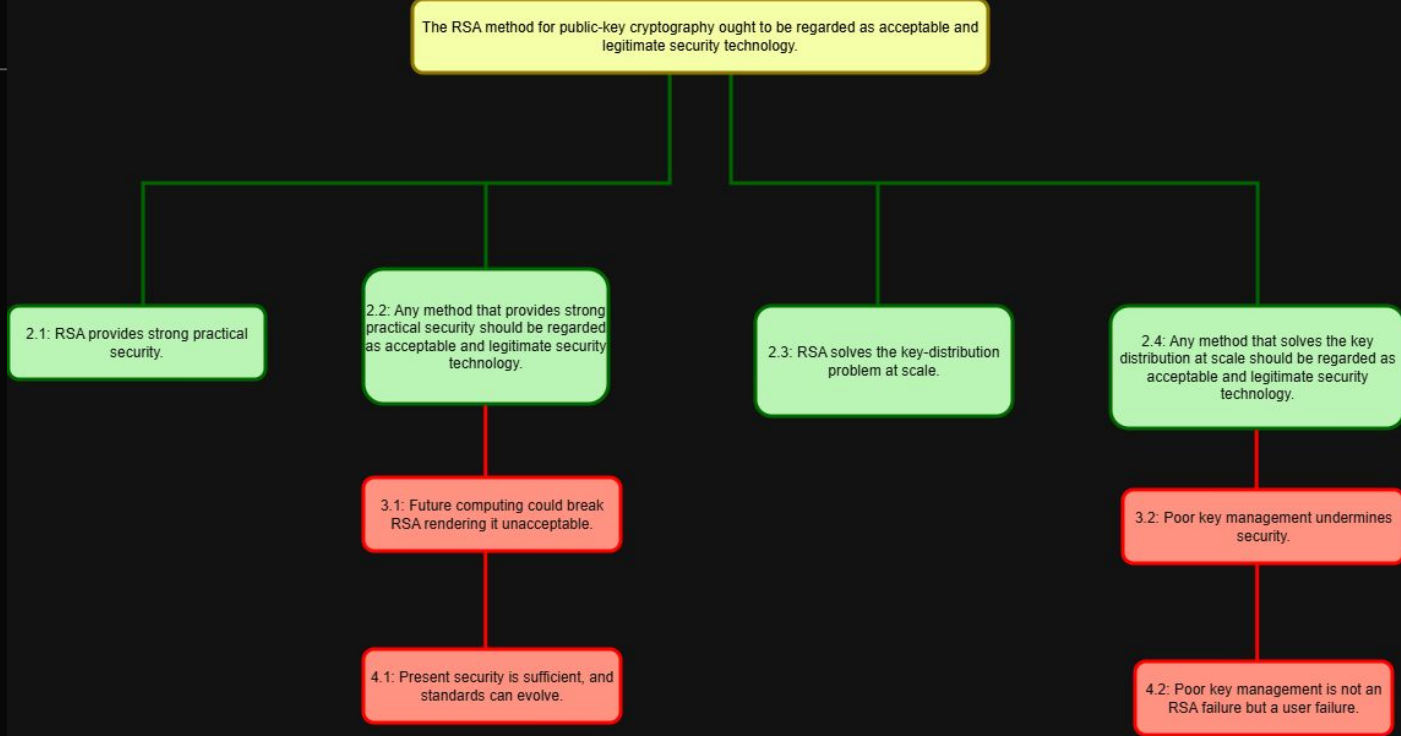
RSA solves the **key-distribution** problem.

Evidence:

- RSA relies on functions that are **easy to compute** but extremely difficult to reverse.
- **Classical encryption requires a secure means**, such as meeting in person, to transmit keys before communication can begin.
- In RSA, a **user can publicly reveal their encryption key** without compromising their secret decryption key
- This allows users to communicate over insecure channels **without ever having met in person** to exchange secrets.



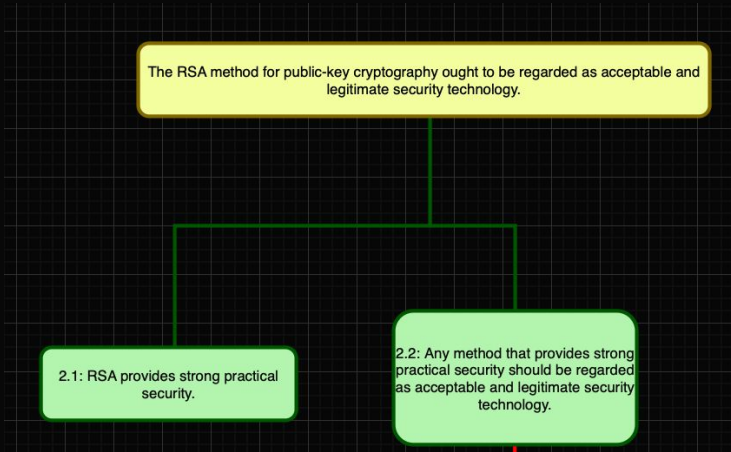
2.2



This argument is valid because the conclusion follows logically from the mathematical premises. However, its soundness depends on the continued truth of the assumption that factoring large numbers is computationally "hard". Given the current state of technology, the RSA method remains the most promising implementation for secure electronic communication.

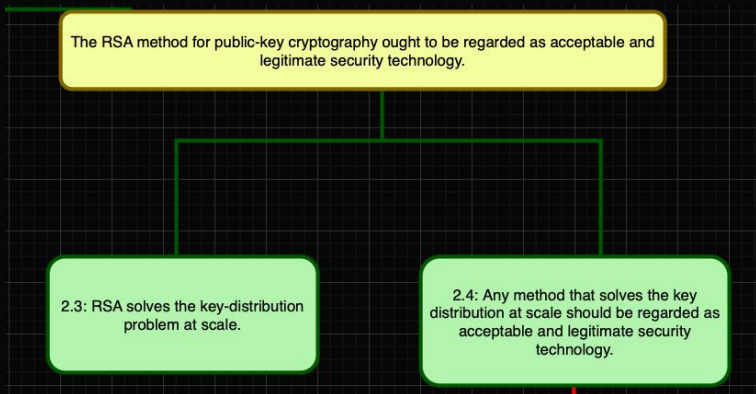
2.3 Find Missing Co-Premises

If we agree RSA provides strong practical security, we must also agree that any method providing such security should be considered acceptable.

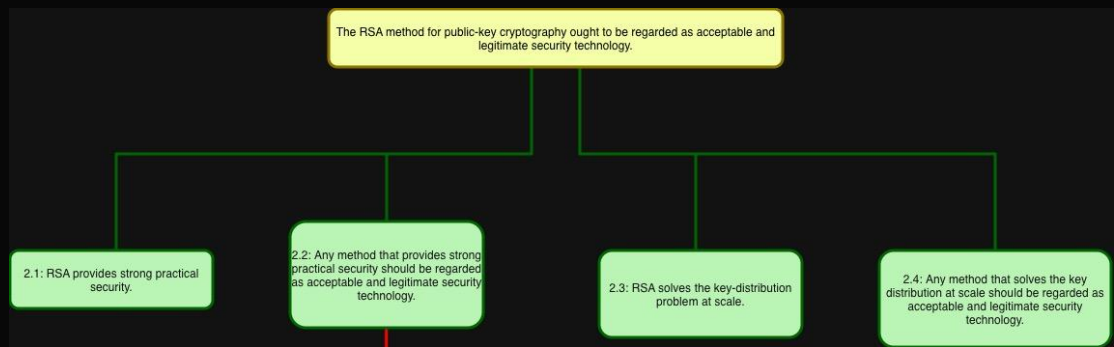


2.3 Find Missing Co-Premises

If RSA solves the key-distribution problem at scale, and we believe solving that problem is a requirement for legitimate technology, then the claim is proven.



2.3 Find Missing Co-Premises



Unit 3: Handle Objections

3.1: Identify Objections

3.2: Develop Rebuttals

3.3: Evaluate the Strength of the Argument

3.Extra: Discarded Arguments

3.1 Objection One - Future computing could break RSA, rendering it unacceptable.

- Premise assumes the RSA method has long-term strength; permanence
- Security based on difficulty of factoring n ; innovation can decrease difficulty
- "No techniques exist to prove that an encryption scheme is secure."

The RSA method for public-key cryptography ought to be regarded as acceptable and legitimate security technology.

2.2: Any method that provides strong practical security should be regarded as acceptable and legitimate security technology.

3.1: Future computing could break RSA rendering it unacceptable.

3.2 Rebuttal One - Present security is sufficient, and standards can evolve.

- Objection targeted the strength of the method against future innovation
- In the present moment, it is effective, and RSA is difficult to decipher due to the random unique n
- For the future, innovations on the security side can improve methods

The RSA method for public-key cryptography ought to be regarded as acceptable and legitimate security technology.

2.2: Any method that provides strong practical security should be regarded as acceptable and legitimate security technology.

3.1: Future computing could break RSA rendering it unacceptable.

4.1: Present security is sufficient, and standards can evolve.

3.1 Objection Two - Poor key management undermines security.

- Premise focuses on the fact that RSA can solve the key distribution problem
- Good key distribution is pointless if the keys are managed poorly
- Security flaws increase with poor key management

The RSA method for public-key cryptography ought to be regarded as acceptable and legitimate security technology.

2.4: Any method that solves the key distribution at scale should be regarded as acceptable and legitimate security technology.

3.2: Poor key management undermines security.

3.2 Rebuttal Two - Poor key management is not an RSA failure but a user failure.

- Objection targeted distribution effectiveness against management issues
- Management isn't a problem of the method, but of users
- Distribution of keys gives access, security of keys varies (like passwords)

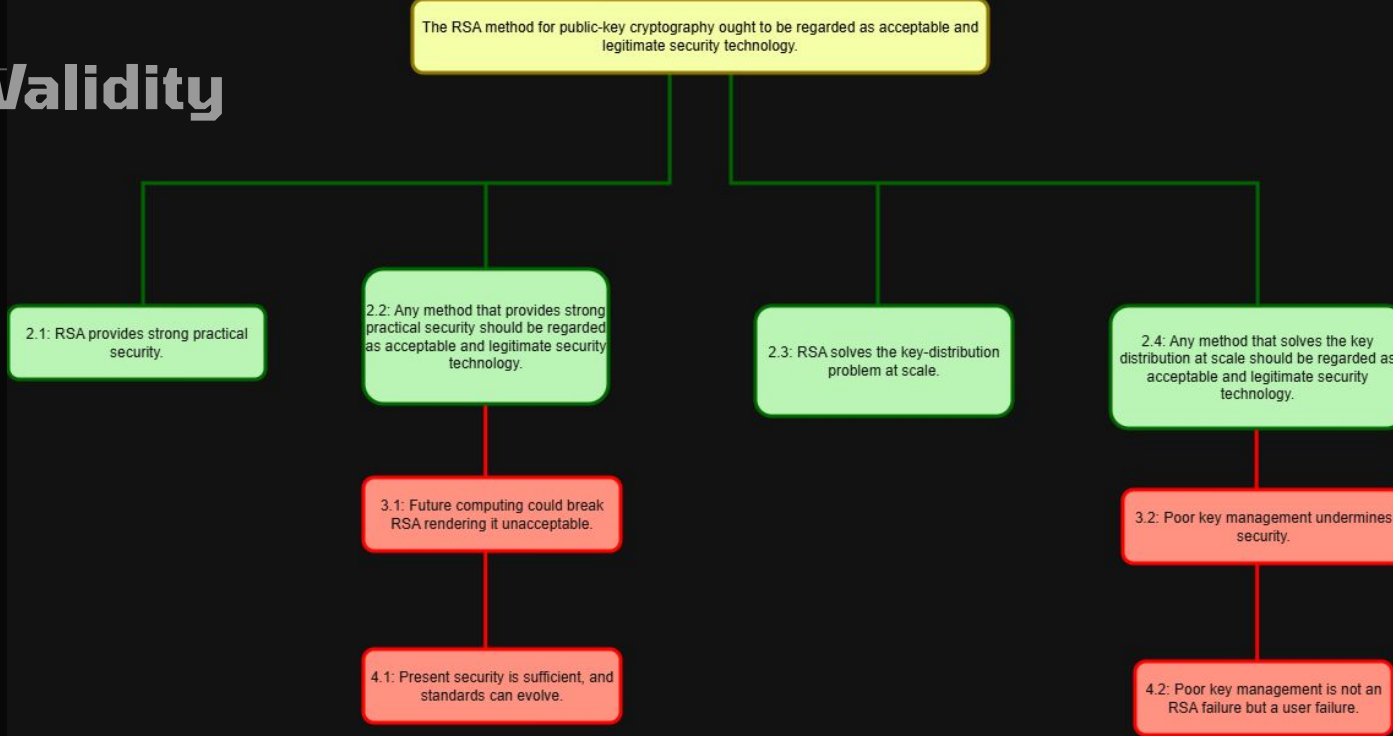
The RSA method for public-key cryptography ought to be regarded as acceptable and legitimate security technology.

2.4: Any method that solves the key distribution at scale should be regarded as acceptable and legitimate security technology.

3.2: Poor key management undermines security.

4.2: Poor key management is not an RSA failure but a user failure.

3.3 Validity



This argument is valid because the conclusion follows logically from the mathematical premises. However, its soundness depends on the continued truth of the assumption that factoring large numbers is computationally "hard". Given the current state of technology, the RSA method remains the most promising implementation for secure electronic communication.

3.3 Rebuttal One - Note

- Hypotheticals about the future are hard to dispute
- The rebuttal mitigates the argument by saying that, though computing can advance to break it, it could also advance to improve it
- Almost like a "So What?"

The RSA method for public-key cryptography ought to be regarded as acceptable and legitimate security technology.

2.2: Any method that provides strong practical security should be regarded as acceptable and legitimate security technology.

3.1: Future computing could break RSA rendering it unacceptable.

4.1: Present security is sufficient, and standards can evolve.

Arguments Considered with Older Premises

Objection to... "This method is an easy to implement asymmetric public-key cryptosystem."

Objection: "The method has a longer computation time."

Rebuttal: "Modern technology can compensate for the time difference."

Objection to... "Asymmetric encryption circumvents the security key distribution problem."

Objection: "What happens if keys are compromised?"

Rebuttal: "New keys can be generated quickly."

Arguments Considered with Older Premises

Objection to... "The method is as secure as factoring large numbers."

Objection: "Could the private keys be determined without factoring?"

Rebuttal: "The key must be factored since it is only defined by the modulus of its primes."

Objection to... "Factoring large numbers is very difficult."

Objection: "There are methods to factor large numbers."

Rebuttal: "Those methods are still difficult to implement."

Unit 4: Assess Soundness

4.1: Assess All Claims as True or False

4.2: Assess for Soundness

4.3: Write Your Evaluation

4.1 Are Our Premises True?

Q: Does RSA truly eliminate the need for couriers?

A: Yes, the public-key architecture mathematically ensures this.

Q: Is factoring actually "hard"?

A: Yes, but this may not be the case in the future.*

4.2 Assessing Soundness

The RSA argument is logically **valid**: the conclusion follows from the premises, and there are no structural issues with the argument

The RSA argument is **contingently sound**: it is absolutely true that public keys eliminate the key distribution problem, and at present it is true that factorization is hard enough to be secure

4.3 Final Evaluation

The argument presented by Rivest, Shamir, and Adleman is logically valid, as the conclusion follows from the premises and there are no underlying structural issues with the argument. However, its soundness depends on the continued truth of the assumption that factoring large numbers is computationally "hard". Given the current state of technology and mathematics, the RSA method remains the most promising implementation for undeniably secure electronic communication, but this may change in the future. Therefore, I would declare this argument as presently sound, and bearing no further burden of proof.

Questions?

THANK YOU!

Bibliography

Rivest, R.L., Shamir, A., and Adleman, L. (1978). A Method for Obtaining Digital Signatures and Public-Key Cryptosystems. Communications of the ACM, 21(2), 120-126.